

CFD Report

Summary

This report presents the use of Computational Fluid Dynamics (CFD) to evaluate the momentum, heat and mass transfer in a jacketed tubular plug flow reactor (PFR). The model was developed using the Finite Element Method (FEM) in COMSOL. The system simulates the exothermic hydrolysis of propylene oxide (PO) into propylene glycol (PG), using the given operating conditions and thermophysical properties specified in the brief.

The PFR was modelled in 2D axisymmetric along with heat balance, transport of diluted species and turbulent flow interfaces. The given diffusivity value of $1 \times 10^{-9} \text{ m}^2/\text{s}$ was used to set up the mass balance equations, because the value obtained from Molecular Dynamics (MD) is big compared to the experimental data. The base case was constructed using the $k - \omega$ turbulent model with a 'Fine' mesh setting, as a result of a mesh independence study. The velocity, temperature and concentration profiles were plotted as a function of three axial position (inlet, 50% length, outlet), illustrating high temperature and concentration gradients near the wall at 50% length and outlet.

The outlet velocity profile can be described by the $u^+ = C \ln y^+ + D$, with calculated constants $C = 4.09$ and $D = -1.20$. Additional properties, including wall shear stress, pressure gradient, friction velocity, friction distance, fanning friction factor and thicknesses of the viscous sub-layer and the buffer layer, were calculated as shown in Table 3. Reducing flowrate and increasing in the inlet coolant temperature are proven to be effective strategies to increase the concentration of PG at the outlet. The $k - \omega$ model, though slower, demonstrated much better convergence and suitability than $k - \epsilon$.

Method and Setup

Turbulent flow produces eddies and small oscillations at spatial scales, which is not feasible to be resolved with the ordinary Navier-Stokes equations (Frei, 2017). Hence, this model used the Reynolds-Averaged Navier-Stokes (RANS) to minimise computational cost while maintaining modelling accuracy. The two-equation RANS models are explored:

- **Base Case:** The $k - \omega$ was chosen as the main model as it is known to be better at evaluating the flow in the viscous sub-layer near the wall than $k - \epsilon$, which is important for this model. It is also well suited for curvature flows such as the flow through a pipe bend (Frei, 2017).
- **Comparative study:** $k - \epsilon$ is known to require less computational power and running time, and its robustness made it a popular choice for industrial uses.

Part 1: Computational Domain and Mesh Generation

Geometry

The tubular plug flow reactor (PFR) was modelled in **2D-axisymmetric**, representing the cross section of the PFR in the r - z plane, presenting both radial and axial gradients around the central axis. The PFR is modelled as a vertical cylindrical pipe with a length of 1.5 m, and a radius of 0.2 m ($L = 1.5 \text{ m}$, $R = 0.2 \text{ m}$). This PFR is assumed to be mounted horizontally to neglect the gravitational effect.

Mesh Strategy and Independence Study

Mesh Strategy: the physics-controlled mesh was chosen to automatically match the flow physics and generate a boundary layer mesh near the wall. The mesh is dense near the reactor walls to resolve the high velocity and temperature gradients expected within the viscous sublayer in the turbulent flow. The bulk domain utilised the triangular mesh to save computational power in the bulk region with lower flow concentration gradients, as shown in Figure 10 in the **Appendix**.

Mesh Independence Study: A grid convergence study was conducted to assess the stability of the model, the dependency on the mesh sizes, and find the best mesh sizes for this model. Four mesh levels were tested – Coarse, Normal, Fine and Extra Fine, evaluating the effect on the critical numerical results – The centreline velocity ($u_{max,outlet}$), pressure drop (ΔP), outlet conversion ($X_{PO,out}$) and outlet temperature (T_{out}) and also running time and physical memory required as summarised in Table 1.

Table 1. Mesh independence study results

Mesh Size	Running time (s)	Physical Memory (GB)	ΔP (Pa)	T_{out} (K)	$X_{PO,out}$	$u_{max,outlet}$ (m/s)
Coarse	381	1.50	0.0772	313.44	0.0487	0.0516
Normal	310	2.00	0.0785	313.42	0.0486	0.0510
Fine	281	2.22	0.0754	313.38	0.0485	0.0505
Extra Fine	928	3.68	0.0776	313.38	0.0483	0.0498
Fine vs Extra Fine	-	-	2.9%	0.0%	-0.5%	-1.4%

Surprisingly, the 'Fine' mesh required the shortest run time (281 s), faster than the 'Coarse' and 'Normal'. The 'Coarse' model only runs after increasing the maximum number of iterations from 300 to 1000. This shows that the 'Coarse' and 'Normal' are not sufficient, as they require more iterations (run time) to reach convergence. Comparing the 'Fine' to the 'Extra Fine', the $u_{max,outlet}$, T_{out} , and $X_{PO,out}$ of the two models are differed by less than 2%, indicating stability and grid independence achieved at the 'Fine' level. Though the pressure drop differs by 2.9%, this difference is only 0.0022 Pa. As the overall pressure drop of about 0.07-0.08 Pa is very low, due to the small reactor dimensions and inlet flow, this difference is negligible. As a result, 'Fine' is the most suitable mesh size for the base case.

Part 2: Turbulent Flow and Steady State (base case)

The base case was developed using the $k - \omega$ turbulence model with the 'Fine' mesh size, assuming the flow is turbulent and fully developed. The mass balance was solved with the transport of diluted species interface, energy balance with heat transfer in fluids (ht), and the momentum with the turbulent flow $k - \omega$ model. These interfaces were coupled and solved simultaneously with the **Multiphysics Couplings: Nonisothermal Flow** links the turbulence flow with the heat transfer, while **Reacting Flow** links the turbulence flow with the transport of diluted species. This approach adds the effect of turbulent eddies to the heat transfer and effect of turbulent fluctuations on the conversion or reaction.

Boundary Conditions

The boundaries were defined around the centreline on the r-z plane of the axisymmetric at inlet ($z = 0$), outlet ($z = L$) and wall ($r = R$). The inlet flow was assumed as fully developed turbulent. The outlet was set to have standard gauge pressure, and the automatic wall treatment of the $k - \omega$ interface was utilised to automatically construct the fine mesh near the wall and coarser mesh in the bulk. Boundary conditions of the base model are summarised in Table 2.

Table 2. Boundary Conditions

Interface	Boundary Location	Boundary Condition and Explanation
Momentum Balance (Turbulent Flow, k-w)	Inlet ($z = 0$)	Fully developed turbulent flow, based on average velocity calculated from the given average flowrate $u_{in} = Q_{in}/(\pi R^2)$
	Outlet ($z = L$)	Pressure is set as standard outflow $P_{out} = 0 \text{ Pa}$
	Wall ($r = R$)	No slip, automatic wall treatment
Energy Balance (Heat Transfer in Fluids)	Inlet ($z = 0$)	Prescribed inlet temperature, $T_{in} = 40 \text{ }^\circ\text{C}$
	Outlet ($z = L$)	Convective outflow
	Wall ($r = R$)	Convective heat flux $U_c (T - T_{in,c})$, where U_c is given
Mass Balance (Transport of Diluted Species)	Inlet ($z = 0$)	Use prescribed concentration of water and propylene oxide
	Outlet ($z = L$)	Convection dominate transport, no restriction on the outlet concentration
	Wall ($r = R$)	No content can leave the reactor, No flux at the wall

Assumption

The model developed followed the assumptions provided in the coursework brief, including negligible change in volume due to mixing, negligible gravitational effect (horizontally mounted PFR), 2D axisymmetric, and no catalyst contribution in the reaction. The following additional assumptions were made to further simplify the model and save computational power:

- Isotropic turbulence assumes the fluctuations are the same in all directions, the accuracy retained with the implementation of $k - \omega$ wall treatment function.
- The diffusion coefficient of propylene oxide is the same as propylene glycol, which is valid as turbulent eddy dominates the mass transport.
- The inlet flow is a fully developed steady state incompressible turbulent flow.

Results and Analysis of Base Case

Velocity gradient is the same along the three axial locations of the PFR as shown in Figure 1. The profile is flat up to 0.14 meters and gets steeper near the wall. This demonstrates uniform flow along the PFR and the effectiveness of the turbulent mixing due to the identical gradients of the three axial positions. Same as the velocity profile, the temperature profile at the outlet and at the half-length positions demonstrates a sharp gradient closer to the wall or the cooling jacket. Figure 2 also shows that the temperature gets higher along the pipe, demonstrating the reduction in the effect of the cooling jacket.

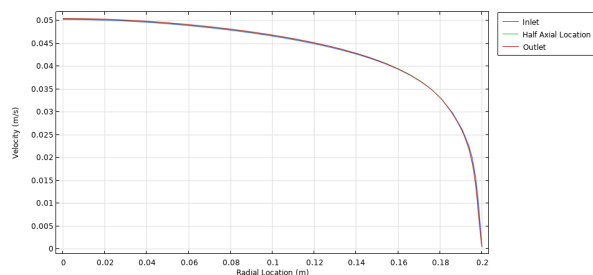


Figure 1. Radial velocity profiles

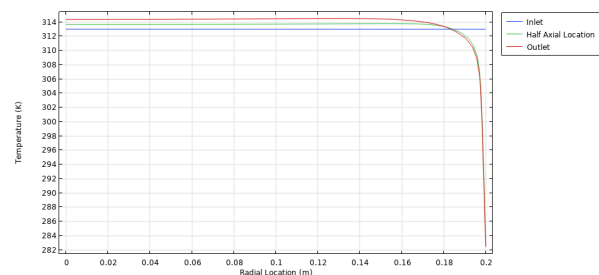


Figure 2. Radial temperature profiles

The concentration profiles show the relationship between the production of PG and the consumption of PO in **Error! Reference source not found..** There is no formation of PG at the inlet as expected, and the PO concentration is constant at the input (1600 mol/m^3). The main observation is the dominance effect of no-slip over the cooling near the wall, as the concentration of PG increases and is highest closer to the wall at the half axial and outlet locations. This indicates that the residence time may have more effect on the PG production over the coolant temperature, which will be assessed in the later section. The results also show that the production of PG is relatively low, as the concentration of PG at the outlet core is at least 10 times smaller than the concentration of PO at the inlet.

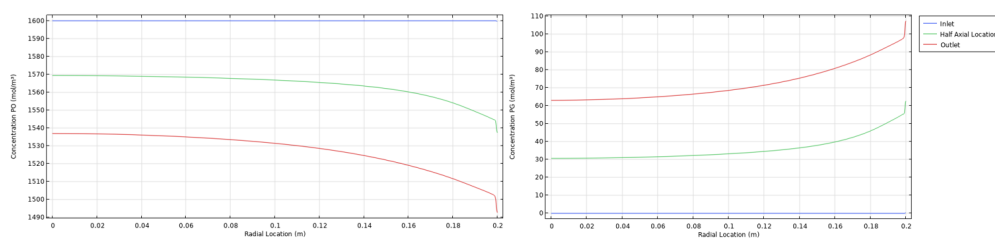


Figure 3. Radial concentration profiles: Propylene Oxide (left), Propylene Glycol (Right)

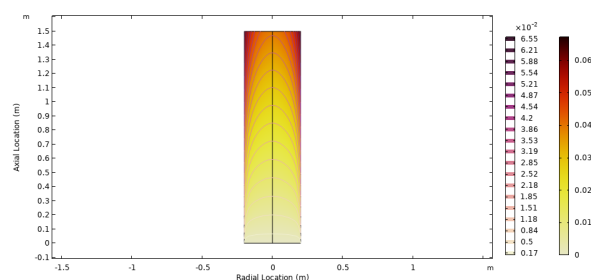


Figure 4. Conversion of Propylene Oxide in the reactor

Compare with the $k - \varepsilon$ model

It took only 185 seconds and 1.76 GB physical memory to simulate the $k - \varepsilon$ model with the same setting mesh level (Fine) as the base model (281 seconds, 2.22 GB). Moreover, the $k - \varepsilon$ model required only 82 iterations to reach convergence, compared to 115 iterations for the $k - \omega$ model. These results proved that $k - \varepsilon$ is more efficient, and hence more widely used industrially. However, as demonstrated by the convergence plots below, the $k - \varepsilon$ plot demonstrated significant fluctuations and oscillations when comparing to the $k - \omega$ convergence, which is more stable at the iterations between 30 and 60. The sudden drop in error of the $k - \omega$ convergence at the 60 iterations demonstrates the transition between the viscous sub-layer and turbulent core. These proves of stability and accuracy of the $k - \omega$ model is why it is more suitable solve to the turbulent system of the PFR, with a sharp gradient near the wall.

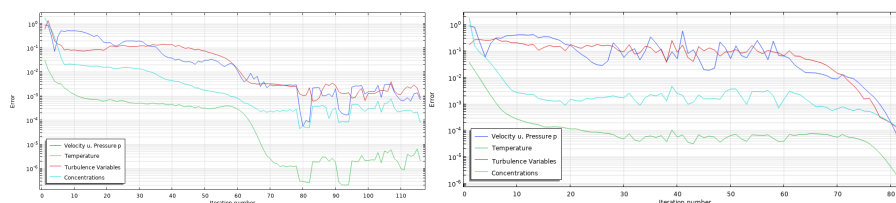


Figure 5. Convergence plot: $k - \omega$ (left), $k - \varepsilon$ (right)

Proved that the velocity profile at the outlet can be described by $u^+ = C \ln y^+ + D$

Figure 6 shows that the velocity profile at the outlet of the base model can be described by a logarithmic velocity profile for $0.018 \text{ m} < y < 0.05 \text{ m}$, corresponds to a dimensionless wall distance of $53 < y^+ < 151$. The data was exported from COMSOL, and curved fitted using Python, ($C = 4.09$ and $D = -1.20$). The full methods in the **Appendix A**. The calculated slope C is steeper than the turbulent core value of 2.5 (lecture note), this represents the cooling effect at the outlet near the wall at $0.018 \text{ m} < y < 0.05 \text{ m}$. Assuming the velocity profile generated can be described by the universal velocity profile, the further values, including wall shear stress, pressure gradient, friction distance, fanning friction factor and thicknesses of the viscous sub-layer and the buffer layer are calculated as shown in **Table 3** in the **Appendix**.

Two strategies that can increase the concentration of the PG at the outlet.

Flowrate can be reduced to increase residence time and therefore conversion. As shown in Figure 7, reducing the flowrate from 0.005 to $0.003 \text{ m}^3/\text{s}$, the concentration of PG at the outlet centre increased from approximately 63 to 105 mol/m^3 , which is more than 66% compared to the base case (**Error! Reference source not found.**). The second strategy is to increase the inlet coolant temperature. Increasing the coolant temperature from 4°C to 14°C reduced the cooling rate, led to better conversion near the wall and also better distribution of the concentration of PG along the reactor as shown in Figure 8 and Figure 9 in the **Appendix**.

Limitations of the CFD model

The main limitation is the computational power available. COMSOL is a powerful finite element tool that has a 3D modelling option, but solving the Navier-Stokes equations in 3D requires a lot of computational power and processing time. The utilisation of the 2D axisymmetric approach simplified the model and significantly reduced the set-up and convergence time. However, this assumes perfect radial symmetry, causing the loss of 3D flow disturbances that could potentially affect the mixing and conversion rate. For instance, the swirl due to the upstream pumps and vortices due to the gravitational effect are being neglected from this model. Though the PFR is mounted horizontally, the gravitational effect can still induce some buoyancy.

The $k - \omega$ model and the 'Fine' mesh are very accurate and suitable for the computational power available. However, finer mesh sizes and other RANS methods (such as $k - \omega$ SST) and other advanced methods, including the Large Eddy Simulation (LES) can also be explored to accurately evaluate the gradient near the wall and flow behaviour if a better computer is available. Lastly, this PFR is simple with no curvature or backflow, therefore, there is no requirement for such high complexity for this study purpose, in which the COMSOL $k - \omega$ model is very effective.

Reference

Frei, W., 2017. Which Turbulence Model Should I Choose for My CFD Application? COMSOL [Online]. Available from: <https://www.comsol.com/blogs/which-turbulence-model-should-choose-cfd-application> [Accessed 30 November 2025].

Appendix

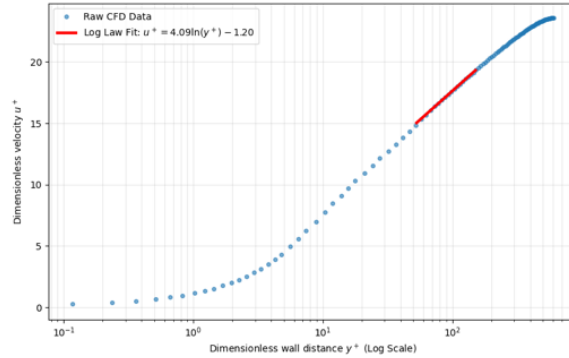


Figure 6. Plot of u^+ vs y^+ (Log Scale) at the outlet, use to fine C and D constants (Python)

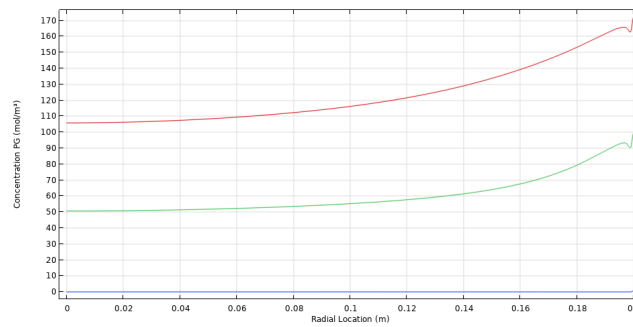


Figure 7. Concentration profile of PG at the flowrate of $0.003 \text{ m}^3/\text{s}$

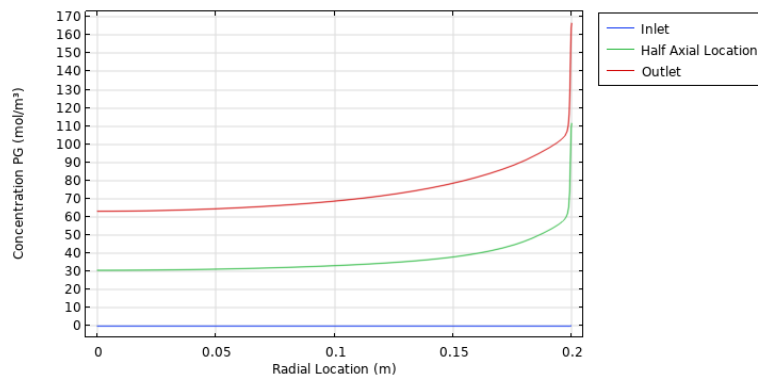


Figure 8. Concentration profile of PG, coolant temperature of 14°C

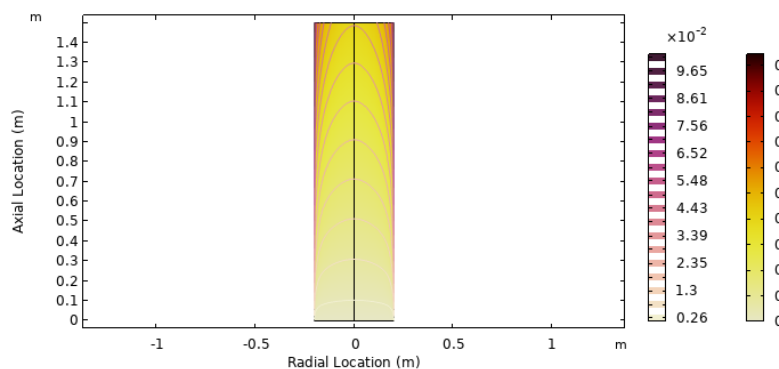


Figure 9. Conversion of Propylene Oxide in the reactor, coolant temperature of 14°C

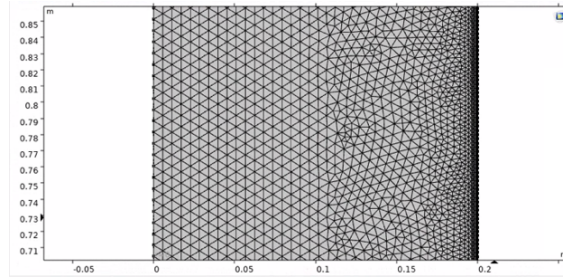


Figure 10. Mesh distribution, Fine size k-w

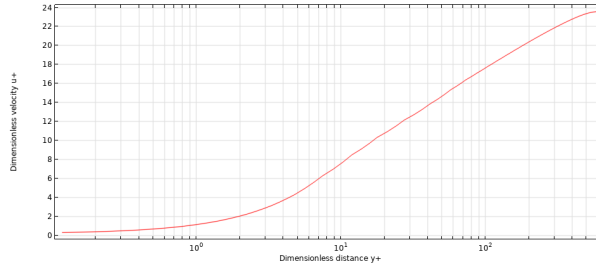


Figure 11. Plot of u^+ vs y^+ (Log Scale) at the outlet (COMSOL)

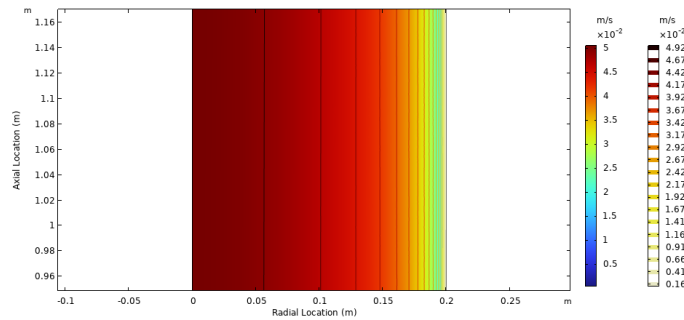


Figure 12. Radial velocity profile of the base $k - \omega$ model, contour + surface

Appendix A

Dimensionless u^+ calculations

The friction velocity was obtained at the outlet wall as $u^* = 2.14 \times 10^{-3}$ m/s using point valuation.

The dimensionless u^+ was calculated using the equation: $u^+ = u/u^*$

Dimensionless y^+ calculations

y^+ was calculated using the model density and dynamic viscosity to find kinematic viscosity and then use $y^+ = yu^*/\nu$ and $\nu = \mu/\rho$.

y is a distance from the wall, so $y = R - r$.

Where ν is the kinematic viscosity, η is the system dynamic viscosity, and ρ is the system density.

Example conversion from 0.018 m $< y < 0.05$ m to 53 $< y^+ < 151$.

$$y^+ = \frac{\rho_{y=0.018} 0.018 u^*}{\mu_{y=0.018}} = 53$$

$$y^+ = \frac{\rho_{y=0.05} 0.05 u^*}{\mu_{y=0.05}} = 151$$

Appendix B

Further values calculated assume the velocity profile can be described by the universal velocity profile, the result is summarised in the Table 3.

Table 3. Further values calculated assume the velocity profile can be described by the universal velocity profile

	Values and units
Friction velocity, u^*	$2.14 \times 10^{-3} \text{ m/s}$
Wall shear stress, τ_0	$4.58 \times 10^{-3} \text{ Pa}$
Viscous sub-layer thickness, δ_{VSL}	$2.81 \times 10^{-3} \text{ m}$
Buffer layer thickness, δ_{Buffer}	$1.68 \times 10^{-3} \text{ m}$
Friction distance, y^*	$5.62 \times 10^{-4} \text{ m}$
Fanning friction factor, f	5.78×10^{-3}
Pressure gradient, $\Delta P/L$	0.0503 Pa/m

Wall shear stress, τ_0

The u^* and ρ obtained from the point evaluation at the wall outlet as: $u^* = 2.14 \times 10^{-3} \text{ m/s}$ $\rho = 1000 \text{ kg/m}^3$

Using $u^* = \sqrt{\tau_0/\rho}$ $\rightarrow \tau_0 = u^{*2} \times \rho = (2.14 \times 10^{-3} \text{ m/s})^2 \times 1000 \text{ kg/m}^3 = 4.58 \times 10^{-3} \text{ Pa}$

Viscous sub-layer thickness, δ_{VSL}

Using $y^+ = yu^*/\nu$ or $y^+ = y\rho u^*/\mu$

As viscous sub-layer only extended to $y^+ = 5$, using the dynamic viscosity at the outlet wall from point evaluation $\mu = 1.20 \times 10^{-3} \text{ Pa s}$

$$\therefore \delta_{VSL} = y_{VSL} = \frac{5\mu}{\rho u^*} = \frac{5 \times (1.20 \times 10^{-3} \text{ Pa s})}{1000 \text{ kg/m}^3 \times 2.14 \times 10^{-3} \text{ m/s}} = 2.81 \times 10^{-3} \text{ m}$$

Buffer layer thickness, δ_{Buffer}

Using the same approach as the viscous sub-layer thickness calculation but at $y^+ = 30$

$$\therefore \delta_{Buffer} = y_{Buffer} = \frac{30\mu}{\rho u^*} = \frac{30 \times (1.20 \times 10^{-3} \text{ Pa s})}{1000 \text{ kg/m}^3 \times 2.14 \times 10^{-3} \text{ m/s}} = 1.68 \times 10^{-2} \text{ m}$$

Friction distance, y^*

As $y^* = \mu/\sqrt{\tau_0\rho}$

$$\therefore y^* = \frac{1.20 \times 10^{-3} \text{ Pa s}}{\sqrt{4.58 \times 10^{-3} \text{ Pa} \times 1000 \text{ kg/m}^3}} = 5.62 \times 10^{-4} \text{ m}$$

Fanning friction factor, f

The general equation for the fanning factor and the average velocity at the outlet, u_{avg} from **line average** is used. $u_{avg} = 3.98 \times 10^{-2} \text{ m/s}$

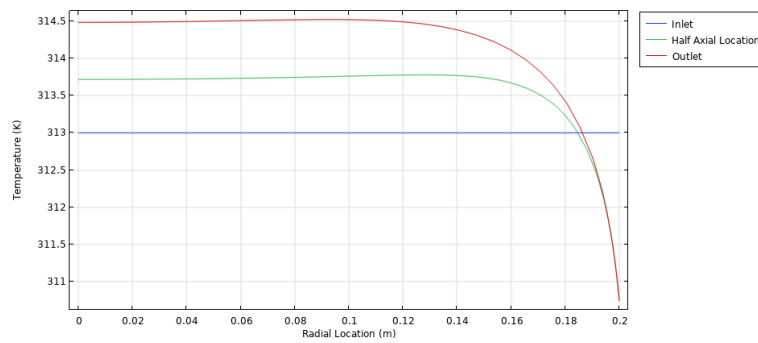
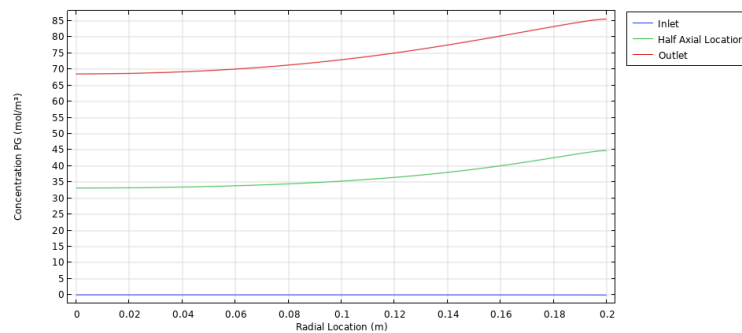
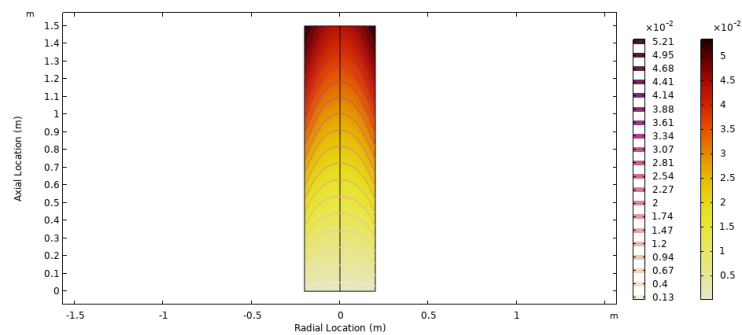
$$\therefore f = \frac{2\tau_0}{\rho u_{avg}^2} = \frac{2 \times (4.58 \times 10^{-3} \text{ Pa})}{1000 \text{ kg/m}^3 \times (3.98 \times 10^{-2} \text{ m/s})^2} = 5.78 \times 10^{-3}$$

Pressure gradient, $\Delta P/L$

Pressure gradient is assumed to be change in pressure over the length of the reactor:

The pressure drop is obtained earlier in the mesh validation section as $\Delta P = 0.0754 \text{ Pa}$

Hence, $\Delta P/L = 0.0754 \text{ Pa}/1.5 \text{ m} = 0.0503 \text{ Pa/m}$

Additional $k - \epsilon$ resultsFigure 13. Radial temperature profiles ($k - \epsilon$)Figure 14. Radial concentration of Propylene Glycol ($k - \epsilon$)Figure 15. Conversion of Propylene Oxide in the reactor ($k - \epsilon$)**Operating and thermophysical properties and nomenclatures**

Properties	Values	Unit
Activation energy, E_a	75.36	kJ/mol
Frequency factor, A	1.70×10^{13}	1/h
Overall heat-transfer coefficient, U_c	1300	W/m²/K
Thermal conductivity mixture k_m	0.56	W/m/K
Inlet temperature, T_{in}	313	K
Inlet temperature of the coolant, $T_{in,c}$	277	K
Heat of reaction, ΔH_{rx}	-84.67	kJ/mol
Total flow rate, Q_{in}	0.005	m³/s
Average flow velocity, u_{in}	$u_{in} = Q_{in}/(\pi R^2)$	$u_{in} = Q_{in}/(\pi R^2)$
Propylene oxide concentration, inlet $C_{in,PO}$	1600	mol/m³

Water concentration, inlet C_{in,H_2O}	48000	mol/m ³
Reactor radius, R	0.2	m
Reactor length, L	1.5	m
Molar weight, propylene oxide, M_{PO}	58.095	g/mol
Molar weight, water, M_{H_2O}	18	g/mol
Molar weight, propylene glycol, M_{PG}	76.095	g/mol
Density, propylene oxide ρ_{PO}	830	kg/m ³
Density, water ρ_{H_2O}	1000	kg/m ³
Density, propylene glycol ρ_{PG}	1040	kg/m ³
Dynamic viscosity of mixture, μ	0.001	kg/m/s
Reference temperature T_{ref}	293	K
Molar specific heat capacity, PO	75.5	J/mol/K
Molar specific heat capacity, water	145.6	J/mol/K
Specific heat capacity of the mixture in the inlet $C_{p,in}$	$(c_{pm_H_2O} \cdot c_{H_2O0})/\rho_{H_2O} + (c_{pm_poxide} \cdot c_{poxide0})/\rho_{poxide}$	$(c_{pm_H_2O} \cdot c_{H_2O0})/\rho_{H_2O} + (c_{pm_poxide} \cdot c_{poxide0})/\rho_{poxide}$
Thermal conductivity of mixture	0.56	W/m/K
Specific heat capacity of coolant	4180	J/ kg K